Biswajit Jana

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EDUCATION

University of Hertfordshire Hatfield, United Kingdom
MSc Astrophysics with Advanced Research, Commendation 2022–2024

- Dissertation: *Closed-Loop Feedback Control System for the Exoplanet High-Resolution RV Spectrograph (EXOhSPEC) Development*
- Relevant modules: Star Formation and Evolution; Foundations of Cosmology; Physics of Astronomical Spectra; Early Universe and Galaxy Formation; High Energy Astrophysics; Galaxy Structure and Evolution; Statistics and Analysis; Research Techniques in Astrophysics; From Stars to Planets.

University of Engineering and Management Jaipur, India
BTech Electronics and Communication Engineering, First Class with Distinction 2017–2021

- Dissertation: *Dimming Controlled Visible Light Communication System using Raspberry Pi*
- Thesis work published in *Optical and Quantum Electronics*, Springer.

PUBLICATIONS

- **Jana, B.**, et al. "Implementation of Dimming Controlled Visible Light Communication using Raspberry Pi." *Optical and Quantum Electronics*, 53, 725, 2021. DOI: 10.1007/s11082-021-03362-4.
- **Jana, B.**, et al. "GSM Controlled Location Specific Garbage Collecting Smart Bin." *9th Annual Information Technology, Electromechanical and Microelectronics Conference*, Jaipur, India, 2019. DOI: 10.1109/IEMECONX.2019.8877007.
- **Jana, B.**, et al. "Priority Encoder Using Reversible Logic Gates in QCA." *8th IEEE Annual Information Technology, Electronics and Mobile Communication Conference*, Vancouver, Canada, 2017, pp. 319–323. DOI: 10.1109/IEMCON.2017.8117242.

RESEARCH AND PROJECT EXPERIENCE

Closed-Loop Feedback Control System for EXOhSPEC 2023–Present
MSc research project, University of Hertfordshire

- Developed a closed-loop stabilisation framework for EXOhSPEC, a high-resolution radial-velocity spectrograph testbed for exoplanet instrumentation.
- Integrated IDS3010 interferometric sensing, Meerstetter TEC control, BME environmental sensors, MaxIm DL centroid tracking, and adaptive-optics correction into Python-based feedback workflows.
- Quantified environmental coupling: 1 hPa pressure change produced ∼0.4 μm optical-path-length variation, while 1°C temperature change produced ∼6 μm variation.
- Achieved ∼±0.3°C enclosure stability over multi-week operation, later reaching millikelvin-level TEC stability during closed-loop experiments.
- Developed Optical Path Length and pixel-drift correction models using environmental data, interferometry, and guider centroid errors; diagnostics were implemented through Grafana, InfluxDB, serial logging, and web monitoring.
- Integrated AO as a fine stabiliser: Stage-1 tests validated ∼0.15 px/step response and reduced mean post-correction residuals from ∼0.44 px to ∼0.15 px; selected hybrid runs reduced dY RMS from ∼2.72 px to ∼0.69 px.

Gaia DR3 H–R Diagram Laboratory 2026–Present
Independent scientific software and browser-based stellar-population visualisation

- Built a live browser-based Gaia DR3 laboratory for exploring stellar populations on the Hertzsprung–Russell diagram using real catalogue data.
- Implemented 10+ interactive components: H–R diagram view, stellar-region overlays, colour mapping, Sun reference marker, local 3D Galactic projection, telemetry panels, live loading console, zoom controls, export tools, and progressive browser-side data loading.
- Solved practical frontend-science problems including large CSV handling, browser memory limits, rendering performance, progressive data loading, and scientific interface clarity.
- Skills developed: Gaia catalogue handling, browser-side scientific visualisation, JavaScript data pipelines, public documentation, GitHub Pages deployment, and research communication.

- Links: GitHub | Interactive Tool

ExoLight Transit Lab and Exoplanet Visualisation Tools 2026–Present

Independent browser-based exoplanet simulation and science-communication project

- Developed a live exoplanet transit visualisation tool for explaining transit geometry, normalised light curves, flux dips, limb darkening, atmospheric transmission, and possible exomoon-style signatures.
- Built a modular static web application with interactive parameters, local target data, scientific diagrams, blog-linked documentation, and public GitHub deployment.
- Added quantitative science outputs including transit-depth visualisation, normalised flux curves, ingress/egress interpretation, planet–star radius-ratio explanation, and duration/geometry-based transit communication.
- Skills developed: transit photometry modelling, scientific frontend design, modular JavaScript architecture, GitHub Pages deployment, and public-facing astrophysics writing.
- Links: GitHub | Live Project | Blog

Detection of Pulsars by Image Stacking in Large FITS Datasets

Pulsar image-stacking and large astronomical image-processing project

- Addressed time and memory constraints in processing large astronomical FITS-image datasets.
- Applied the Binapprox algorithm for efficient median estimation across 9,587 astronomical images in FITS format.
- Developed image-stacking and median-combination workflows to improve faint-source detectability, with emphasis on efficient computation, FITS handling, and signal extraction for pulsar-search style datasets.

Dimming Controlled Visible Light Communication System using Raspberry Pi 2019–2021

BTech final-year project, University of Engineering and Management, India

- Implemented a Multi-Header Hybrid Pulse Position Modulation based visible-light-communication system using Raspberry Pi and low-cost commodity hardware.
- Evaluated dimming levels of 0.25, 0.5, and 0.75; achieved communication distances of ~3 m at dimming level 0.25 and ~3.5 m at dimming levels 0.5 and 0.75.
- Observed improved throughput behaviour across varying incidence angles; work later published in *Optical and Quantum Electronics*.

ADDITIONAL ASTRONOMY RESEARCH EXPERIENCE

Case Studies in Astrophysics University of Hertfordshire

MSc research-based coursework Sep–Dec 2022

- **Extra-solar planets:** analysed transit dips and radial-velocity variations using DACE, Python, and `lightkurve` to estimate planetary parameters and interpret exoplanet systems.
- **Brown dwarfs:** used literature review and model-fitting arguments to classify a target as a brown dwarf with estimated temperature ~1100 K and silicate atmospheric features; proposed further analysis using PICASO 3.0 and trimmed atmospheric models.
- **Hunting for Death Stars in Gaia:** investigated close stellar encounters and identified GJ710 as a potential close-passage candidate, including perihelion-distance and uncertainty analysis.

Bayfordbury Observatory Practical Experience University of Hertfordshire

Observational astronomy, calibration and public outreach 2022–2024

- Gained practical experience in CCD calibration, photometry, image acquisition, guiding, and spectroscopic data reduction using observatory facilities including Meade telescopes and associated imaging systems.
- Participated in planetarium demonstrations and public telescope sessions, explaining observational techniques to students and visitors.

INTERNSHIPS, SUMMER AND WINTER PROGRAMMES

LUMI SPACE / SEPnet Summer Placement United Kingdom

Evaluating Microbolometer Camera Performance for Satellite Laser Ranging 5 Aug–4 Sep 2024

- Evaluated microbolometer-camera performance for satellite laser ranging and orbital-debris tracking; constructed link-budget comparisons for microbolometer, CMOS, CCD, and quad-detector systems.
- Modelled detector responsivity, beam divergence, atmospheric transmission, refractive-index effects, and performance

limits; proposed improvement routes using high-TCR materials and optical-antenna coupling.

Summer Research Project University of Hertfordshire
Identifying Halo T Dwarf Candidates using VISTA, DES, and WISE Survey Data Jun–Jul 2023

- Worked under Professor David Pinfield to analyse VISTA, DES, and WISE survey data for potential Halo T dwarf candidates.
- Contributed to candidate assessment, follow-up planning, and a CNTAC telescope-proposal submission for Magellan/FIRE spectroscopic follow-up observations in 2025A.

Carl Sagan Exoplanet Summer Workshop Caltech / NASA Exoplanet Science Institute
Atmospheric modelling and exoplanet analysis training 2023

- Gained hands-on exposure to exoplanet atmospheric modelling, brown-dwarf atmosphere interpretation, and retrieval tools including PICASO and petitRADTRANS.

Radio Astronomy Summer School Gauribidanur Radio Observatory / Raman Research Institute / Naxxatra
Two-element radio interferometer and solar-transit observation Feb–May 2022

- Worked under Dr. Ramesh Balasubramanyam; demonstrated a two-element radio interferometer and observed solar transits, gaining practical exposure to radio astronomy instrumentation and interferometry.

Society for Space Education Research and Development Online
Optical–radio catalogue cross-matching for elliptical-galaxy classification 11 Oct–21 Nov 2020

- Worked under Mr. Sundar M. N. to cross-match optical and radio catalogues and explore machine-learning-based classification using multi-wavelength astronomical data.

IIT Kharagpur Winter Internship India
Electronics and robotics research-oriented projects 22 Dec 2019–23 Jan 2020

- Worked under Professor Sudip Misra on low-cost high-frequency detecting circuits using FPGA and various other tools and calibration/synchronisation of multiple drones for swarm configuration.

Foxmula / Inversion Consultancy LLP India
Analysis of Diseases and Symptoms using Python 27 Aug–27 Sep 2019

- Developed applied Python data-analysis skills using scientific programming libraries in an introductory disease/symptom analysis project.

SCIENTIFIC BLOGS, SIMULATIONS AND PUBLIC SOFTWARE

- Project blog and portfolio: biswajit1999.github.io/Biswajit_Jana.github.io/blog/.
- Topics include Gaia DR3 visualisation, exoplanet transits, radial velocity, control systems, gravitational waves, pulsars, cosmology, and scientific visualisation; skills developed include JavaScript simulation design, GitHub deployment, technical documentation, diagram design, and public research communication.

TALKS AND POSTER PRESENTATIONS

- *Testing Performance of Microbolometer Detector for Satellite Laser Ranging Application*. SEPnet Poster Presentation, London, UK, 20th November 2024.
- *High Resolution RV Spectrographs: ANDES and PID Loop Implementation in EXOhSPEC*. University of Hertfordshire, 7th March 2024. [Slides]
- *Optimising Path Length Stability in Laser Interferometers using Air Refractive Index*. University of Hertfordshire, 8th February 2024. [Slides]
- *Advancement in Precision: Laser Interferometer Control System*. University of Hertfordshire, 8th December 2023. [Slides]
- *State of the Art: Radial Velocity Spectrograph*. University of Hertfordshire, 9th November 2023. [Slides]

TECHNICAL TRAINING AND SELECTED COURSES

- **9 Robots in 8 Days, MYWBUT Winter Training, 2017–2018:** built PC-controlled, photovore, edge-detecting, obstacle-mapping, line-follower, differential-speed, gesture-controlled, and sumo robots using Arduino and sensors.
- **MATLAB Fundamentals with Arduino Integration, MYWBUT Summer Training, 2018–2019:** used MATLAB

and Simulink for scientific programming, engineering problem solving, and basic control-system circuit design.

- **Internet of Things, MYWBUT Summer Training, 2019–2020:** developed IoT architecture using HTML, CSS, PHP, NodeMCU ESP8266, sensors, and a home-automation project.
- **Selected courses:** First Order Optical System Design; Data-driven Astronomy; Astrobiology and the Search for Extraterrestrial Life; Astronomy: Exploring Time and Space; AstroTech; Basic Course in Astronomy: Introduction to Stars.

TECHNICAL SKILLS

Programming and analysis: Python, JavaScript, HTML, CSS, MATLAB, C/C++, PHP, SQL, Bash *(beginner–intermediate working proficiency*
Astronomy and data analysis: NumPy, SciPy, pandas, Matplotlib, Astropy, `lightkurve`, DACE, FITS workflows, Jupyter Notebook, Aperture Photometry Tool, RSpec.
Instrumentation and optics: laser optics, interferometry, IDS3010 displacement sensing, radial-velocity spectrograph stabilisation, environmental control, optical alignment, adaptive optics correction, PID control.
Telemetry and control systems: Grafana, InfluxDB, serial communication, RS-232/RS-485 communication, sensor fusion, real-time logging, feedback-control dashboards.
Hardware and embedded systems: Arduino, Raspberry Pi, NodeMCU ESP8266, PSoC 5LP, embedded boards, IoT systems, visible-light-communication prototypes.
Software and tools: MaxIm DL, MATLAB, Simulink, QCA Designer, PSpice, PSoC Creator, Git, LaTeX, Overleaf.
Scientific communication: technical reports, research posters, telescope-proposal preparation, scientific blogging, public research visualisation, outreach presentations.

REFERENCES

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